

International Islamic University Chittagong
Faculty of Shariah and Islamic Studies
Department of Quranic Sciences and Islamic Studies

Semester End Examination: Spring -2026

Program: BA (Hon's)

Course Code: STAT-3503

Course Title: Statistics for Research in Sharia'h Studies.

Time: 2.5 Hours

Full Marks: 50

[Answer the following questions. Figures in the right margin indicate full marks.]

QN	Description of Questions	Marks	CLOs & PLOs	Cognitive learning														
1	a Analyze the effectiveness of the range as a measure of dispersion and compare it with other measures of variability in statistical decision-making	2	CLO2, PLO3	Understand														
	b Assume that, your pathology lab provides the following details of blood test carried out on 100 patients for diabetes.	8																
	<table border="1"> <tr> <td>Blood sugar</td><td>90-100</td><td>100-110</td><td>110-120</td><td>120-130</td><td>130-140</td><td>140-150</td></tr> <tr> <td>No. of patients</td><td>16</td><td>20</td><td>22</td><td>25</td><td>10</td><td>4</td></tr> </table> Calculate mode deviation from the mean blood sugar.	Blood sugar	90-100	100-110	110-120	120-130	130-140	140-150	No. of patients	16	20	22	25	10	4			
Blood sugar	90-100	100-110	110-120	120-130	130-140	140-150												
No. of patients	16	20	22	25	10	4												
	Or																	
1	a Analyze the objectives of measuring variation in statistics.	3	CLO2, PLO3	Understand														
	b Explain the properties of an ideal measure of variation and justify which measure is most appropriate for statistical analysis.	5+2		Understand														
2	Total revenue: $TR(Q) = 250Q - 3Q^2$ Total cost: $TC(Q) = Q^2 + 50Q + 400$ (i) Write down the profit function. (ii) Find $d\pi/dQ$, set it equal to zero and solve for Q^*. (iii) Calculate Total Revenue (TR) at $Q=Q^*$ (iv) Calculate Total Cost (TC) at $Q=Q^*$ (v) Calculate Profit (π) at $Q=Q^*$ and verify it equals $TR-TC$ from your previous steps.	(2+3+1.5+1.5+2)	CLO3, PLO4	Apply														
	Or																	
2	Differentiate the following functions using first-order differentiation (i) $d/dx (x^2 + 3x^4 + 4x^5)$ (ii) $d/dx (2\sqrt{x})$ (iii) $d/dx (3x^2 + 1)(x^3 + 2x)$	(2*5)=10	CLO3, PLO4	Apply														

$$\begin{array}{r}
 16 \\
 20 \\
 \hline
 36 \\
 22 \\
 \hline
 58
 \end{array}$$

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		(iv) $d/dx[(x^2-a)(x^2-b)]$ (v) $d/dx\left(\frac{x^2-3x}{2x^2+4x}\right)$															
3	a	Explain how sampling improves the efficiency of statistical research.	5	CLO3 .PLO4	Evaluation												
	b	Describe the principal steps involved in designing and conducting a sample survey for a research thesis, highlighting how each step contributes to the reliability and validity of the findings.	5														
4	a	Critically evaluate the importance of studying Econometrics for students of the Shariah Faculty, particularly in analyzing Islamic economic issues, supporting evidence-based research, and improving policy formulation.	6	CLO4 .PLO4	Analyze												
	b	Critically evaluate the limitations of Econometrics and assess how these limitations may affect the reliability of economic analysis and policy decisions.	4														
5		A researcher wants to find out if there is any relationship between age of the husbands and the ages of wives. He took a random sample of 5 couples whose respective ages are given below. <table border="1"> <tr> <td>Age of husbands</td><td>25</td><td>20</td><td>26</td><td>32</td><td>37</td></tr> <tr> <td>Age of wives</td><td>18</td><td>20</td><td>21</td><td>21</td><td>30</td></tr> </table> (i) Compute the regression equation of age of wives on husbands. (ii) Compute the regression equation of husbands on age of wives. (iii) Determine the most likely age of wife for age of husband 36 year. (iv) Determine the most likely age of husband for age of wife 35 year. (v) Compute the correlation of coefficient.	Age of husbands	25	20	26	32	37	Age of wives	18	20	21	21	30	(3+3+1.5+1.5+1)	CLO4 .PLO4	Apply
Age of husbands	25	20	26	32	37												
Age of wives	18	20	21	21	30												

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